Incredible Edible Cell

Purpose: You will use several different food items representing the various organelles or cell structures found in plant and animal cells to construct an edible cell model. This activity will help you learn more about the different parts of a cell and their functions.

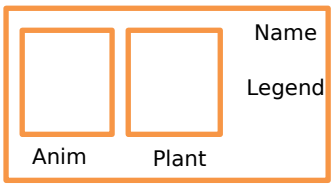
Background:
All cells have a *cell membrane*, which forms a barrier to separate the cell from its
environment. The cell membrane surrounds a gel-like fluid called *cytoplasm*, which is the medium that holds all the organelles of the cell. The large *nucleus* of a cell is similar to your brain because it functions as the cell's control centre. The nucleus contains genetic material that is used as instructions for directing cell functions. *Endoplasmic reticulum* (ER) surrounds the nucleus and helps to form and move proteins throughout the cell.

Attached to the ER are *ribosomes*, which produce proteins and can also be found floating elsewhere in the cytoplasm. The "powerhouses" of the cell are the *mitochondria*, which convert food energy to usable energy. Water, food, and other materials are stored in *vacuoles*. *Chloroplasts* are present only in pant cells and are the site for photosynthesis.

Materials: Milk Arrowroot biscuit, Milk Coffee biscuit, icing, a pop stick, 2x jaffas, 4 jelly beans, 2 x green tic tac or mini m&m, 6 small strips of liquorice, some 100's & 1000's.

Procedures:
Do NOT eat any of the materials unless given permission by your instructor!! Use gloves when handling food items!

1. Draw a table into your lab notebook, completing the table by filling in the main function of each organelle – nucleus, cell membrane, mitochondria, cytoplasm, chloroplast, endoplasmic reticulum, ribosomes.
2. Label the edge of a piece of paper with your name and each cell type (animal or plant), and a legend indicating which lolly will represent each organelle



3. If you are making an *animal* cell, use a milk arrowroot base. If you are making a *plant* cell, use a milk coffee base
4. Spread frosting evenly over the base to represent the cytoplasm.

5. Assemble your model by sticking the appropriate “organelles” into the “cytoplasm,” showing correct spatial relationships between different types of organelles.

Extension Questions:

1. Given the function of mitochondria, what tissue might contain cells with a high concentration of mitochondria? Why?

2. Why could a cell be called “the functional unit of life?”

3. Why is it important to have a cell membrane when there is a cell wall?

4. How is the nucleus the “command centre” of the cell?

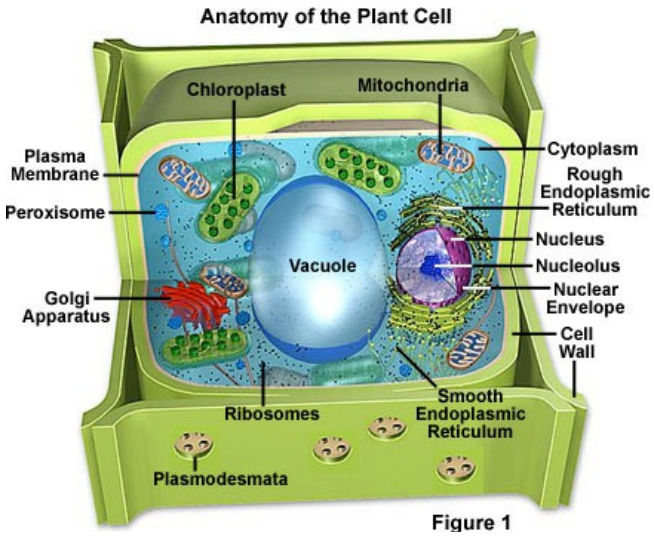
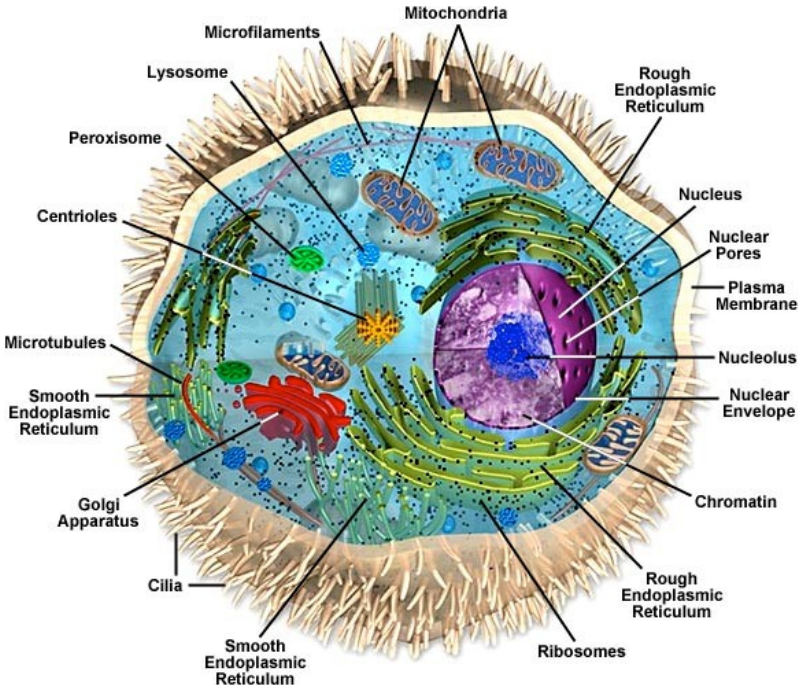


Figure 1


Mitochondria
Microfilaments
Lysosome
Rough Endoplasmic Reticulum
Peroxisome
Nucleus
Centrioles
Nuclear Pores
Plasma Membrane
Microtubules
Nucleolus
Smooth Endoplasmic Reticulum
Nuclear Envelope
Chromatin
Golgi Apparatus
Cilia
Rough Endoplasmic Reticulum
Smooth Endoplasmic Reticulum
Ribosomes